

Interactive Chinese Character Learning in Virtual Reality

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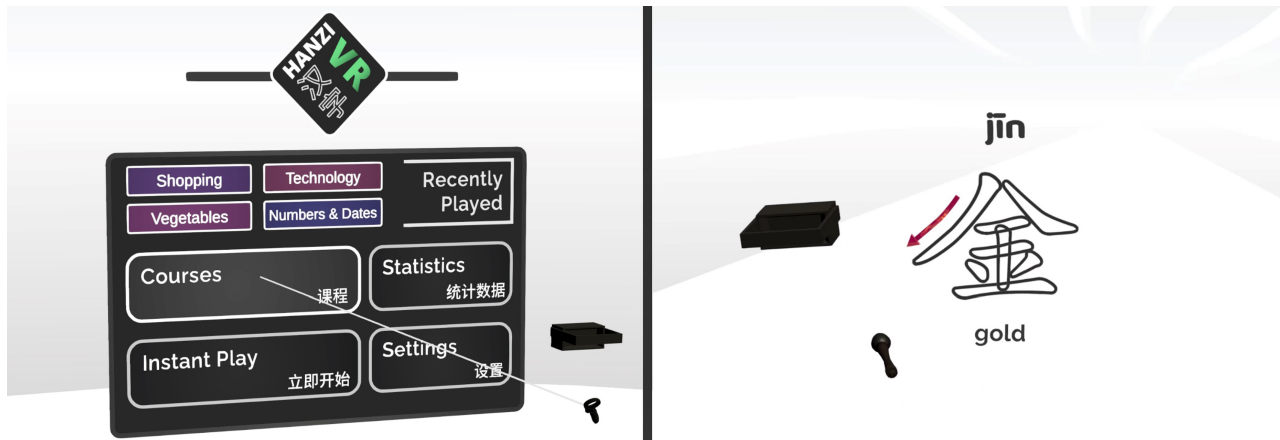


Figure 1: Player using the HanziVR menu (left) and game scene (right) to select and learn Chinese characters.

ABSTRACT

Learning Chinese characters and characters of Chinese origin is a challenging and time-consuming task for native speakers and second language learners alike. Traditional methods to acquire Chinese characters include repeatedly writing the characters on paper or air writing. In scientific research, computer- and mobile-assisted approaches have been implemented that allow users to be interactively guided through the learning process. The recent advancements and availability of virtual reality (VR) headsets with six degrees of freedom tracking and controllers provide an interesting new platform for education applications. The proposed VR application utilizes novel interaction and visualization methods tailored for VR with guided repetition as well as sound and haptic feedback. The augmentation of air writing further enables motor-based learning and allows for the application to be used as an educational exergame.

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1 INTRODUCTION

To help young native speakers and second language learners to acquire unique Chinese characters and those of Chinese origin, different learning methods are used. Traditional methods include the repeated writing of characters on grid paper or the writing on transparent paper over guided progressions of printed Chinese characters, allowing to learn the stroke composition and even stroke order [Zhang 2014]. Another traditional method that is used to learn Chinese characters is air writing, in which the finger is used to trace the character either in the air or on a surface like one's palm or a table [Zhang 2014]. Air writing can be used in combination with the visual imagination of the strokes as well as the pronunciation of the stroke names. Recent research focused on the usage of computer- and mobile-assisted learning methods to support students in learning Chinese characters. Especially applications that use touch input by finger or stylus have been developed and evaluated, with some of the results suggesting that computer- or mobile-assisted learning methods can improve the memory retention of Chinese characters in comparison to traditional methods [Hsiao et al. 2013]. In recent years, the advancements in and availability of virtual reality (VR) technology has also increased the research on the usage and effects of VR for educational purposes [Hartfill et al. 2020; Vazquez et al. 2018].

The proposed VR character learning system and its implementation for the Chinese language, called HanziVR, focuses on extending traditional methods and their advantages through repetition and motor-based representation, while integrating concepts from recent research such as improved visualization, error detection,

gamification, customization and musical sound design. The implemented visualization and augmentation of air writing through VR introduces a novel approach in the research of Chinese character learning applications. It is intended to broadly explore the interaction and design space of VR-based learning applications for Chinese characters and those of Chinese origin. The application also supports the Japanese and Korean language, focusing on the Kanji and Hanja, respectively. The combination of traditional and novel learning approaches is intended to increase memory retention effects, the enjoyment of learning and the experience of flow, over a longer period of use.

2 SYSTEM OVERVIEW

HanziVR is developed with Unity for the Oculus Quest platform. The application allows users to learn Chinese characters in VR by tracing the characters with the Quest controller. HanziVR consists of two scenes, the menu scene and the game scene, as shown in Figure 1. Both scenes have different interaction methods. The menu scene is a collection of 2D screens that can be operated with the controller used as a laser pointer. The game scene has interactable 3D font objects that can be drawn using the controller with a brush metaphor. The controller interaction is additionally supported by haptic feedback in the form of vibrations during different events like hover, press and drawing, as well as auditory feedback, including brush sounds and audio cues for completing or failing strokes.

2.1 Gameplay

The game scene constitutes the major novelty of the VR learning system in comparison to current mobile-assisted systems by extending the learning experience to the third dimension. The user draws the characters stroke by stroke, while the strokes either move towards the user or appear in front of them. After drawing a stroke, it either stays in place or translates back to its original position. The behavior of the strokes is customizable by the user and therefore allows for four different behavior modes. For all modes, a custom error detection implementation decides whether the user has drawn a stroke successfully, that is, in the correct direction and with the targeted speed.

2.2 Customization

For an individualized learning experience, HanziVR supports a variety of customizable elements related to both learning content and gameplay. For an upcoming session, users can select specific vocabulary sets they are interested in, as well as the selection, order and number of presented characters. In addition, users can adjust the difficulty level by changing the game speed to one of five predefined options or a numerical multiplier. Individual fine tuning of character and stroke settings allows users to tailor the experience to their body height and further customize the movements, animations, and positioning required during gameplay.

2.3 Gamification Elements

The most basic gamification element used in HanziVR are points. Points are earned for each stroke drawn and are used as a short-term motivation for the current game session, as they are further used in conjunction with a combo game mechanic. Drawing multiple

correct strokes allows users to increase their combo, which in turn increases the points multiplier. This gamification method is used in many rhythm games, such as *osu!* [ppy 2007], *Beat Saber* [Beat Games 2018] or *Audioshield* [Fitterer 2016]. In HanziVR, the rhythm is not provided by music, but through the direction, length and display duration of single character strokes. For the latter, a custom stroke duration calculation incorporates the order of and distance between incoming strokes during gameplay. Points are further multiplied by the difficulty level of the play session that can be defined by the game speed. As such, the difficulty levels themselves can be seen as gamification elements that motivate users to play on increasingly high game speeds. By creating a state of flow, users might be challenged to consistently perform well at a high difficulty level, and learning the strokes and characters becomes a byproduct of that challenge rather than the challenge itself.

Furthermore, all this data is reflected in the statistics screen that can be considered as an extension to the different gamification elements by providing information about the points, combos, medals and completed courses.

3 CONCLUSION

The presented learning application transfers known concepts from previous research on computer- and mobile-assisted learning systems to VR, leveraging the visualization and interaction capabilities of VR systems. Traditional methods like air writing were enhanced with gamification elements as well as haptic and auditory feedback. The Chinese characters appear as 3D meshes that can be traced using the VR controller. The application includes different configurable settings to customize the learning experience.

From a research perspective, the introduction of a recall mode and its effects on learning success would be interesting aspects to further explore. Moreover, the potential of using similar game mechanics as in HanziVR in a multiplayer environment might be interesting for further research on collaborative and remote learning in VR. Finally, recent advancements of precise hand tracking might offer an interesting opportunity into researching augmented air writing by tracking the finger movement of the user. The hand tracking could also be used in combination with augmented reality glasses, possibly combining the engaging and fun nature of HanziVR with the expected situational flexibility of augmented reality glasses in the future.

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